

CHAPTER 25

FACILITY DESIGN, LAYOUT, AND ORGANIZATION

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After completing this chapter, you will be able to

- identify the aspects and phases of new strength and conditioning facility design;
- identify the aspects of updating or renovating an existing strength and conditioning facility and transitioning an existing building or space for strength and conditioning use;
- explain how to assess the needs of an athletic program to design a facility that is well suited to these needs, including the planning of training spaces to accommodate various size groups of athletes and coaches;
- explain how to design specific facility features, including supervision location, access, ceiling height, flooring, environmental factors, electrical service, mirrors, and storage;
- explain how to arrange equipment in organized groups, creating better traffic flow and enhancing safety throughout the facility; and

- explain the maintenance and cleaning needs for the surfaces and equipment in a strength and conditioning facility.

Organizing the construction and design of a strength and conditioning facility requires a well-thought-out plan created by a committee of experienced professionals. This chapter provides information on the different phases of planning a new facility, along with details about facility design, organization, and equipment maintenance. The chapter also outlines the phases of construction or remodeling of a facility. The design should account for the needs of various stakeholders (e.g., athletes, staff, administrators) for the layout to be most efficient.

General Aspects of New Facility Design

Building a new facility takes significant time and planning. One of the first things to do is assemble a committee of professionals. The committee should consist of a builder, architect, designers, lawyers, financial departments, and people who will be involved in using the facility. At least one strength and conditioning professional, or other expert who will be working in the facility, should be included in the committee to provide a perspective that will help maximize the utility and safety of the space. This committee will assist with the facility design as well as consider economic factors of opening a new facility. One of the main challenges in designing a strength and conditioning facility is tailoring the facility to the intended audience. [Figure 25.1](#) outlines the four phases of designing a new facility, along with the main objectives for each phase.

Predesign phase		Design phase	Construction phase		Preoperation phase	
Needs analysis	Feasibility study	Finalize committee	Follow master plan	Arrange equipment	Hire staff	Create cleaning schedule
Master plan	Hire architect	Create blueprint	Check construction progress	Access legal counsel	Assign duties	Create plan for operation

FIGURE 25.1 Phases and objectives of designing a new facility.

Predesign Phase

The **predesign phase** is the first step in building a new facility. This phase should consist of a needs analysis, a feasibility study, and master plan formation. At the end of the predesign phase, a reputable architect should be hired—likely through a bid process—and begin developing a blueprint for the facility.

The **needs analysis** is the step when the designers and experts collaborate to determine the needs of the strength and conditioning program (10). Examples of questions are “How much open space do

you need?” and “What equipment needs to fit into the allotted space?” The needs analysis should be consistent with the specific programming needs of the strength and conditioning and athletic departments. For example, if a strength and conditioning professional regularly incorporates jump testing on force plates during weight room sessions, an area needs to be set aside for that activity. For those serving on the design committee, the needs analysis should convey an effective environment to properly implement the intended program. While site, budget, construction, and other limitations may require a certain level of flexibility as the project progresses, safety considerations for facility users should not be jeopardized (2-7, 9-17, 19-22).

The second part of the predesign phase is called a **feasibility study**. This process should analyze strengths, weaknesses, opportunities, and threats (known as a **SWOT analysis**) (10). The goal of the feasibility study is to ensure that the financial investment will yield a viable and sustainable return. The feasibility study should also evaluate the location, strengths and weaknesses of any proposed ideas, and the growth potential for strength and conditioning and related programs within the department. The potential opportunities and competition threats also need to be considered. This includes market evaluation to identify the best opportunity to train athletes most efficiently, as well as analyzing the target market and growth potential of competing facilities. In the collegiate environment, this may include comparisons to other facilities in the athletic conference or be related to athlete recruiting goals. The aim of the feasibility study is to help determine if the facility will have an opportunity for success against competing programs and businesses.

The **master plan** is the general plan for all phases of the new facility. The master plan should include the building and construction plan, facility design, budget information, and an operational plan to act on once the facility is completed (10). The operational plan should consist of short-term and long-term goals that will enhance

the success of the facility both in the near term and in the future. The operational plan should also include staff development and a plan for the hiring process.

Hiring an architect is the final step of the predesign phase. Selecting an architect is crucial for meeting administration and department goals. It is important to select from a list of architects who have a positive reputation based on previous work. If possible, it is best to choose an architect who has had previous experience in the strength and conditioning industry. The architect will likely be hired through a bid process, meaning that the architect who is hired is often the one who submits the lowest price. Bid plans require efficient use of resources so that the money is spent on what is most useful for the users of the facility.

Design Phase

The **design phase** is the second phase in the process of building a new facility. In the design phase, ideas from the committee come together with regard to the structure and design elements of the facility. It is extremely important to plan around the flow of the facility while adhering to all relevant regulations and codes. Federal, state, and city codes are often highly specific about aspects of the design. City planning departments can provide the local guidelines and applicable codes for constructing a facility.

The first step in the design phase is finalizing the design committee. To reiterate, the committee should include strength and conditioning and other athletic department professionals with experience in facility planning and design. This is also the phase when the designers need to work closely with the architect to produce a blueprint of the facility.

The blueprint and design need to consider the equipment specifications. This will help with the flow of the facility and will influence how effectively people can move from one area of the facility to another once it is operational. Traffic flow is one of the most important aspects of designing the facility. Creating easy

access for people will have the greatest effect on the function and safety of the facility when multiple groups of athletes are present at the same time. The flow of traffic should also allow coaches or supervisors to have a clear view of the entire floor. The locations of any structural pillars, columns, or other obstructions must be considered in the planning of equipment spaces to account for traffic flow and access to emergency exits.

Construction Phase

The **construction phase** is the period of time from the beginning of building the facility until completion. This is usually the longest phase of the process. During construction, the committee needs to continually refer to the master plan to ensure that the project is on track with regard to the established goals and design. Keeping the project on track in relation to the deadline must also become a priority. Delays in construction can add significant financial strain. Having access to legal counsel is beneficial to making sure that construction deadlines are met and extra expenses and legal action are avoided.

Preoperation Phase

The **preoperation phase** consists of the final steps before the facility can open. These steps include finishing the interior decor (note that appearance can greatly enhance the experience of users), aesthetic considerations to support athlete recruiting or membership enrollment, and hiring a staff of qualified professionals. The staff should have at least the minimum required certifications and education. For example, the National Collegiate Athletic Association enforces regulations that require strength and conditioning professionals to be certified by a nationally accredited organization. In addition, many professional sport leagues and tactical positions mandate that a strength and conditioning professional is a Certified Strength and Conditioning Specialist (CSCS) through the National Strength and Conditioning Association (NSCA). Finding hardworking,

knowledgeable, loyal, and trustworthy employees is crucial to running a successful strength and conditioning program. During the preoperation phase, it is important to include a plan for staff development as well. This could vary from semiannual workshops to weekly staff meetings.

During the preoperation phase it is vital to create a plan that will make opening the facility easy. For example, duties such as cleaning and maintenance should be delegated to the staff on a weekly basis. This is easier to do when a cleaning and maintenance schedule has been created and is followed as soon as the facility is operating.

Also during preoperation, a plan for administrative and clerical duties should be created to organize work related to matters such as liability insurance, scheduling, and budgeting. Details, such as choosing a scheduling software package, installing an athlete management system, and delegating clerical duties, should be accounted for before the facility opens.

Existing Strength and Conditioning Facilities

Modifying an existing strength and conditioning facility is similar to designing a new facility, without the process of building from the ground up. A design committee can also be formed for renovating an existing facility, but members such as a builder and architect may not be necessary. An existing facility may also use different hiring processes. Sometimes strength and conditioning professionals continue to work in the same facility regardless of a change of ownership or management. However, there should still be a focus on standards, education, professionalism, and staff development. [Figure 25.2](#) outlines the main steps in modifying an existing facility.

First steps	Form committee	Feasibility study	Needs analysis
Second steps	Create operational plan	Remodel and/or upgrade	Finalize design and decorations
Third steps	Arrange equipment	Create employee duties and schedule	Hire/keep staff

FIGURE 25.2 Steps in modifying an existing facility.

The strength and conditioning professional should assess existing equipment based on the current and future needs of everyone who uses the facility.

Satellite Training Facilities

It has become common in recent years for college and professional sport organizations to add alternative and sport-specific satellite training facilities to accommodate the needs of growing strength and conditioning departments. Other examples of this include providing a strength and conditioning facility for visiting team use, as is mandated by some professional sport leagues, or designing a studio space for individualized or small group training sessions. Special attention should be given when planning satellite training facilities, especially if existing, dated, or nontraining spaces are being converted for strength and conditioning use.

Satellite training facilities may benefit from smaller pieces of equipment based on the size of the space available (e.g., half rack vs. full rack). In small facilities, it can be beneficial to install multipurpose pieces of equipment rather than equipment with a single function (9). A satellite training facility that does not have dedicated walkways separating the equipment areas should be set up with equipment bolted and secured around the perimeter. The center of the room should remain open for transitions between training stations while allowing unimpeded access to emergency exits. As a rule, equipment and safety spacing should never be compromised due to limited space. In accordance with the *NSCA Strength and Conditioning Professional Standards and Guidelines* (16), and as explained in this chapter, a **minimally compliant facility** must be in agreement with mandated building codes and regulations, including accessibility (i.e., wheelchair access and ADA compliance), HVAC, electrical, lighting, flooring, storage, square footage, safety spaces, and overhead clearance and must possess well-maintained equipment and surfaces.

Outdoor Training Spaces

Weight room adjacent or satellite outdoor training spaces have become more common in geographically moderate climates. In addition, the global COVID-19 pandemic created increased awareness among strength and conditioning professionals for the benefit of social distancing, including the use of open-air outdoor training spaces for enhanced ventilation (15). There are also scenarios when scheduling or programming can be improved by having strength and conditioning equipment available near sport or practice facilities that do not have a dedicated weight room space. While environmental factors, like weather, must always be considered for athlete safety, this section provides recommendations for the safe use of strength and conditioning equipment in outdoor training areas.

Outdoor training spaces may include multipurpose training lockers or pods and be located next to a game or practice field (e.g., grass or artificial turf) or in proximity to the weight room (e.g., via garage door access). Outdoor training spaces may be for seasonal or year-round use depending on climate. Important considerations include ensuring adequate space for strength and conditioning activity to be performed safely and that standards for equipment, safety spacing, and walkways are clear. Outdoor training surfaces must offer proper traction for athletic activity and should not be wet or damp from recent storms. Ground surface temperatures should also be considered to avoid injury due to excessive heat or burns (e.g., artificial turf may have higher temperatures than surrounding surfaces). Additional considerations should be placed on ensuring appropriate athletic footwear and attire for strength and conditioning sessions held outdoors.

Similar to indoor training spaces, outdoor training spaces should have a posted emergency action plan that is understood by staff and athletes. If training equipment is to be transported outside daily for use, it is advisable for the coaching staff to make this task more

ergonomic by using a dolly, cart, or wagon with sturdy wheels. All strength and conditioning equipment should be inspected regularly, especially if used outdoors, because being exposed to the elements can accelerate wear.

Assessing Athletic Program Needs

One of the more important considerations in building a strength and conditioning facility is the needs and requirements of the athletes and athletic program. The number of athletes, coaching philosophies, athlete ages and experience levels, schedules and availability, and equipment factors can affect the design of the facility (12). When it is time to evaluate such needs, the facility designers should be able to answer these questions:

- *How many athletes will use the facility?* This is an important consideration because the facility size will be greatly influenced by the number of athletes using the facility at the same time (i.e., at peak usage) and the total number who use it per day, week, and season. The number of athletes will also affect the scheduling and flow of the facility if two or more groups are training simultaneously. Based on regulations such as fire codes or other occupational health and safety regulations, the city planning departments will specify the maximum room capacity. In addition to the number of athletes training, it is also essential to consider staffing and supervision. The number of athletes being trained simultaneously will affect the number of strength and conditioning professionals required to safely and effectively monitor the training session (16).
- *What are the training goals for the athletes, coaches, and administration?* Training goals will directly affect equipment selection for the facility. For example, if a strength and conditioning professional wants athletes to focus on plyometrics and agility, the facility will need an area (turf or field) designated for this purpose. The strength and conditioning professional should also use collective training goals of the various stakeholders as a guideline for ordering equipment. If resistance training is to be programmed, racks

and platforms are more space efficient compared to bench or incline bench press setups that serve only one purpose.

- *What are the demographics of the athletes?* The demographics of the users are important when determining the equipment needs of the facility. Are the users primarily older, younger, male, female, middle school, high school, college, or professional? A facility with primarily older populations may not need as much free weight equipment as it does machines. In a collegiate or team environment, use of the equipment may be more efficient if it is grouped into sections so that multiple groups can work out simultaneously without interference.
- *What will the training experience of the athletes be like?* Answering this question will help the strength and conditioning professional devise training plans, which in turn will dictate the equipment needed. Training will vary according to the experience of the athlete. Athletes with no previous resistance training experience may do more bodyweight exercises than advanced athletes, who may do more weightlifting. It is also important to assess whether needs will change over time and whether equipment requirements will vary depending on the season.
- *How will training sessions be scheduled?* The scheduling of athletes or groups is one of the more difficult aspects of running a facility. The number of people on staff, the layout, and the design of the facility can all be affected by the training schedule. If more than one group will be using the facility at one time, it would make sense to divide the facility into sections so the amount of overlap is limited. Ideally, training sessions should be scheduled throughout the day to avoid capacity issues during the busiest or most popular times. This will help maintain suggested staff-to-athlete ratios and ensure that enough equipment is available. (See [chapter 26](#) for additional details.)

- *What technology is needed to administer training as intended?* Planning for technology is important to account for the continued growth of the program. A strong Wi-Fi or Bluetooth connection may be required if the intent is to deliver strength and conditioning programs using technology devices, such as tablets, TV monitors, or cell phone apps. Ensuring access to a strong cellular signal can provide an additional layer of safety in case of an emergency. Additionally, exercise testing and monitoring equipment may require dedicated space to be operated during training sessions and stored securely between uses. This includes specific exercise testing devices, athlete monitoring sensors, and accessories that may require special cleaning, charging, and maintenance.
- *What equipment needs to be repaired or modified?* No equipment in a facility should have issues such as holes, cuts, or missing pieces. If a cable is frayed or damaged, discontinue using the cable column until the cable is replaced. All existing equipment should be repaired or replaced if broken. Existing equipment should be repaired promptly or should be sold to provide funds for new equipment. In some cases, a piece of equipment is in good repair but needs an accessory for an upgrade. For example, cable columns may require new attachments to be fully functional.

Once all of these questions are answered, the process of designing the facility will be much easier. Ideally, the facility designers should have a solution for every problem; this allows for an optimal training environment. In reality, however, budget and space limitations often stand in the way. It is important to focus on what will affect the facility the most and to design from there. For example, not having enough space will have a larger impact on the training environment and scheduling than not having enough of a specialized piece of equipment. The recommended minimum amount of space is 100 square feet (9.3 m²) per individual, including all

athletes, strength and conditioning professionals, and any additional staff (e.g., athletic trainers) or persons present (16).

Designing the Strength and Conditioning Facility

Before arranging and positioning equipment, one must consider the design of the strength and conditioning facility, whether new or existing. The strength and conditioning professional should pay particular attention to facility location and access, structural and functional considerations, environmental factors, and safety and supervision.

Location

The strength and conditioning facility ideally should be located on the ground floor away from offices and classrooms. This will prevent interruptions elsewhere in the building that are caused by dropped weights, music, or other noises common to strength and conditioning facilities. If the facility is not located on the ground floor, the floor must be stable enough to support heavy equipment and equipment that is dropped. The load-bearing capacity should be at least 100 foot-pounds per square foot (4.8 kN/m²) (7).

Supervision Location

An office area for supervision should be in a central location with a clear line of sight and windows or mirrors providing the opportunity to look out and see everyone, including individuals and groups entering or exiting the room. One option is to place the office at an elevation over the floor of the weight room for better viewing.

Access

Every facility should be accessible to people with disabilities, with either a ramp or a wheelchair lift for any change in height exceeding 0.5 inches (1.3 cm). A ramp should run 12 inches for every 1-inch rise (30.5 cm run for every 2.5 cm rise) with a landing at the top and bottom (3), and stairs should have a rough strip on the edge to

help prevent anyone from falling. Another option for persons with disabilities is a mechanical lift or an elevator.

Every facility should have at least one phone that is accessible to persons in wheelchairs. This complies with the Americans With Disabilities Act (3); it also provides safety in the event of an emergency if the only person who can help is in a wheelchair. The phone should be close to the entrance of the facility or near the office area so that a supervisor can call emergency services as soon as possible following an accident (2). The ADA also requires that light switches, water fountains, fire extinguishers, and automated external defibrillator (AED) devices are reachable and that safety signage is viewable by persons in wheelchairs (3).

A weight room should also have double or garage doors so that large equipment and machines can be moved in and out. However, if the hallway or access point is too narrow, the size of the doors will not make a difference (14), because an outside wall will have to be removed temporarily when moving or installing equipment.

Ceiling Height

The ceiling should be high enough to accommodate any jumping or explosive activities. This includes the height of the athlete plus the space required for box jumps, vertical jumps, and weightlifting exercises. A recommended height is 12 to 14 feet (3.7-4.3 m), which should give enough clearance to allow athletes to comfortably perform most activities (14, 18). Certain exercises, such as overhead medicine ball throws, may require additional overhead clearance. In scenarios when sufficient overhead clearance is not available to perform an exercise, using a nearby outdoor field, fieldhouse, or gymnasium should yield enough space. However, these decisions related to session management should be initiated in the planning and design process to account for the impact on additional traffic flow in and out of the facility during training sessions.

Flooring

Several types of flooring options are available for a strength and conditioning facility. The most common option is some type of rubber flooring and an antimicrobial carpet. Another option is indoor turf, which can be useful if the strength and conditioning professional plans to have athletes perform dynamic warm-up, plyometric, agility, or conditioning exercises. Turf is also a good surface for ground-based movements and sled pushes that require a more field-like surface than typical gym flooring. Rubber flooring is available in rolls, tile inserts, and mats and is considered a premier option because it protects the subfloor from heavy weights and provides durable cushioning. Rubber flooring can also be easier to clean than carpet and is generally more durable and long-lasting within a facility. As a general rule for safety, floor surfaces in physical activity areas should be slip-resistant to minimize the risk of exercise and fall-related injuries (2).

For facilities designed to train athletes for weightlifting competitions, platforms should have a middle portion made of wood and outer sections made of rubber, in alignment with competition rules. Especially in high-traffic strength and conditioning facilities, inlaid platforms are a safer alternative to raised platforms because they are flush with the floor and do not present a tripping hazard.

Lighting

Strength and conditioning facilities have special lighting needs to keep athletes safe and focused on training. Key lighting considerations should include energy efficiency and the use of natural lighting when possible (11). To determine proper lighting, one must consider the square footage of the facility, ceiling height, and the amount of natural sunlight entering the facility through windows. Lighting is recommended between 20 to 30 foot candles (fc; 1 fc = lumen/ft²) or 200 to 300 lux (lx; 1 lx = lumen/m²) at floor level for general gym areas and up to 50 foot candles or 500 lux for dedicated resistance training and group exercise areas (2, 11, 18).

The distribution of light throughout the facility is important to avoid shadowy areas or excessive glare. Glare from direct overhead lighting is especially a concern for users lying on horizontal benches or exercise mats (18). Windows can help make a facility feel more open and modern, and mirrors may enhance the amount of light that comes in by reflecting it to spots in the room without natural light (2). Light-emitting diode (LED) lights are a popular choice because of their extended life span and energy efficiency (2). Lights with color temperatures between 4,000° and 5,000° Kelvin are considered best for strength and conditioning facilities because they resemble daylight, provide optimal visibility for enhanced safety, and can have an energizing effect. Finally, adjustable lighting controls, automated on-and-off settings, and motion sensors also can be considered to reduce power consumption, energy costs, and maintenance (11).

Room Temperature, Humidity, and Ventilation

A strength and conditioning facility needs to maintain a comfortable training temperature. The heating, ventilation, and air conditioning (HVAC) system should have the capability to heat and cool individual sections of the facility. Most sources indicate that between 68 °F and 78 °F (20-26 °C) is an appropriate temperature range, with many suggesting an optimal range of 68 °F to 74 °F (20-23 °C) (2, 5, 18). If the facility is too hot or too cold, the athletes and staff may be uncomfortable and the quality of training will be affected.

Relative humidity should also be monitored in the strength and conditioning facility (6). In any space where there is physical activity, relative humidity should not exceed 60%. This helps to prevent bacterial and microbial growth, avoiding the spread of infections and disease (2, 15). In addition, areas in a facility with excessive moisture can contribute to rust and damage weight room equipment (5). This may be more of a concern if the weight training facility is located near a humid pool or spa area within the building. Ensuring

that any adjacent pool or spa areas are equipped with negative exhaust air flow helps to prevent moisture and chlorine odor from escaping into other areas of the training facility (2).

Ventilation and circulation are primary concerns with regard to facility design and the HVAC system. The HVAC system, a fresh air exchange system, and ceiling or box fans should provide necessary ventilation and circulation that are in accordance with ANSI/ASHRAE 62.1 requirements for health clubs and weight rooms (6). Air systems must provide air circulation at proper levels according to the needs of specific equipment areas and occupant loads (6). For example, aerobic training areas require a higher circulation rate than resistance training spaces (2). Room air should be exchanged a minimum of 8 to 10 times per hour—optimally 12 to 15 times per hour to prevent odors and stagnant air (5). If fans are used, a general guideline is two to four fans for every 1,200 square feet (111 m²). Fans help keep the air from feeling stagnant and muggy, which makes for a better training environment.

Sound Systems and Noise Control

While not essential, many facilities have sound systems that help create a motivating training environment conducive to hard work. The two main considerations for installing a sound system are speaker volume and placement. The sound volume should be less than 90 decibels so that athletes can hear instructions and coaching cues (2). To avoid damage, speakers should be elevated above the height of all training equipment. Placing speakers in the corner allows sound to project toward the middle of the room so that it is more evenly distributed. The use of headphones is not recommended. Headphones can interfere with coaching and instruction and limit athletes' awareness of their surroundings in the case of an emergency.

Background noise and excess external noise are other issues to consider. Such sounds can emanate from sources ranging from the HVAC system to the people in the next building. In a facility that has

yoga or dance classes, sound-absorbing material should be used in the floors and walls to prevent music from reaching others in the facility. Absorbent rubber flooring can be used to reduce noise caused by people jumping, running, or dropping things.

Electrical Service and Devices

A strength and conditioning facility typically requires more outlets than other buildings. Some outlets may require relatively higher voltage for more powerful equipment, such as stair climbers, elliptical trainers, and treadmills. The electrical service also needs to be properly grounded to protect the system from lightning strikes or power surges. Ground fault circuit interrupter (GFCI) outlets are also necessary to ensure safety in the event of an electrical short (1).

With the increase of technology now being used in weight rooms, the placement of wall, floor, and elevated power outlets should be considered to limit future issues with devices and power cords once the facility becomes operational. Specifically, power cords should never extend into exercise areas, safety spacing, walkways, or spotter areas. In addition, any equipment and device charging stations should be in an office or storage area away from weight room traffic. If televisions will be used in the facility, the use of elevated outlets and secured wall or ceiling mounts are recommended. Planning ahead for the use of technology is valuable, especially if the intent is to use television monitors or overhead projectors to communicate workouts and announcements.

Walls and Mirrors

Mirrors can be used for many purposes in a strength and conditioning facility. They can be used as a coaching tool if racks and platforms are placed in the right position, because they provide immediate visual feedback for improved spatial awareness and safety (14). Mirrors can also be used to enhance a room aesthetically.

Mirrors should be placed at least 6 inches (15 cm) away from any equipment (e.g., behind a rack of dumbbells) and a minimum of 20 inches (51 cm) above the floor (4). The purpose of the height guideline is to ensure that weights do not roll, bounce, or slide into the bottom of the mirror and cause the glass to break. The standard weight plate is 18 inches (46 cm) in diameter, giving mirrors a 2-inch (5 cm) clearance if the plate is dropped near the mirror or leaned against the wall behind the mirror (14).

Bumper railings or padding may be useful for places that require added protection from damage, such as walls and mirrors. Railings are also used in dance and yoga studios so people can hold on for balance when necessary. Balance railings should be installed securely with safety in mind and checked regularly before any weight-bearing activity. The choice of material and beam structure used for constructing walls (e.g., drywall or cinder block) should be considered for properly securing balance railings and other wall attachments. Also, strength and conditioning professionals who plan to program medicine ball throws into a wall should make sure to use a solid wall, such as cinder block, to prevent damage.

Hydration and Nutrition Spaces in the Weight Room

Drinking fountains and water bottle refill stations are important additions to a strength and conditioning facility. They should be located away from the training area and should not interrupt traffic flow. Water stations and nutrition areas are often placed next to the entrance of the facility or next to the bathrooms and lockers. With the increased emphasis on nutrition and recovery for athletes, nutrition areas that include food preparation and appliances, such as blenders, knives, cutting boards, and refrigeration, are subject to safe food handling requirements. In the United States, such requirements are governed by the U.S. Food and Drug Administration (22), U.S. Department of Agriculture (22), and U.S. Centers for Disease Control and Prevention (CDC) (19), all of which

have policies and recommendations to avoid food poisoning and food contamination. (See Standard 9.1 in the *NSCA Strength and Conditioning Professional Standards and Guidelines* for additional detail on the responsibility of strength and conditioning professionals related to supplements, ergogenic aids, and drugs [16].)

Locker and Changing Room Access

Access to locker rooms is a good benefit within a facility. At minimum, having a shower located somewhere in the building ensures that cleaning up after a workout will be convenient. Good hygiene is important for preventing the spread of infections and disease. Although not everyone will use the showers or lockers, having them available for on-the-go athletes may boost morale and give them another reason to work hard during a workout.

Storage Spaces

A strength and conditioning facility should have a dedicated storage space to keep items such as extra equipment, cleaning supplies, facility maintenance tools, or broken equipment. A larger facility with more equipment requires more storage space than a small facility. Depending on the layout of the facility, or if it has multiple floors, multiple storage areas may be best to avoid excessive clutter or injury from equipment not being used. For example, an athlete or strength and conditioning professional may injure his or her fingers while racking and storing weights on a plate tree in a space that is too narrow and filled with unnecessary equipment accessories. Additional areas to watch for excessive clutter include underneath dumbbell storage racks, around shelving units, in suspension training areas (8), and in corners and other tight spaces. Making sure that equipment is stored safely and securely improves weight room organization and maintains the longevity of the equipment.

In addition to the storage of training equipment, all necessary emergency equipment should be kept in or near the training facility to be quickly accessible (16). Such equipment must include a fire

extinguisher, AED, CPR equipment, and a stocked first aid kit. Additional details about emergency policies and action plans are presented in [chapter 26](#).

Other areas to consider related to storage include determining where athletes will store personal items, such as coats, backpacks, and water bottles, during training sessions. If no locker room is available at the facility, designating an area for personal item storage is important to avoid clutter. The use of cube storage, shelving, or other functional storage solutions can help maintain organizational flow of the weight room.

Arranging Equipment in the Strength and Conditioning Facility

It is essential to consider spacing requirements when arranging equipment in the strength and conditioning facility. A floor plan can be drawn to help with visualizing how to arrange equipment, especially since each type of equipment and mode of exercise comes with its own safety and efficiency recommendations.

Equipment Placement

Equipment should be grouped into sections such as a stretching and warm-up area, agility and plyometrics, free weights, aerobic area, and resistance machines. Ideally, free weights and racks should be organized along the wall, and there should be walkways between the free weights and machines (1). Racks and machines also can be placed in rows with the back of the equipment facing each other. This allows for improved flow through the weight room, preventing congestion and maximizing usage space.

Using line of sight as a guideline, shorter equipment is better placed in the middle of the room so that strength and conditioning professionals and staff can easily see across the facility. Planning ahead and organizing the equipment so it can be used as a circuit or exercise pairing will also help with traffic flow. Tall machines should be bolted to the floor, a column, or the wall to prevent tipping over.

Cardiovascular machines should be in their own section and should be lined up and organized such that the treadmills, elliptical trainers, stair climbers, ski trainers, rowers, and bikes are grouped together. Similarly, dedicated exercise and performance testing devices should be in their own section and grouped to ensure the consistent administration, validity, and reliability of testing procedures. Keep in mind that most machines requiring electricity will have to be along a wall unless electrical outlets are available in the floor or on a column. Equipment should also be placed away

from the walkway to minimize the chances of someone tripping over it or on associated electrical cords.

Racks, barbells, and dumbbells should have a minimum of 3 feet (0.9 m) of space between equipment to allow for movement without danger to the athlete or spotter. If a spotter or multiple spotters are needed (e.g., on a rack), providing additional space may be required.

Traffic Flow

Traffic flow is heavily influenced by equipment placement. Many facilities are one big room, and walkways can be created via arrangement of the equipment. As mentioned, most racks or machines are best placed lined up in a row running the length of the facility. This usually creates two or three main walkways that should be 5 to 7 feet (1.5-2.1 m) wide (2, 3, 13). For safety, all individuals in the training facility should have clear access to an exit at all times. Ensuring proper equipment and walkway spacing is important for allowing clearance to move freely throughout the facility and for ADA-required wheelchair access (3, 13).

Weight Room Pillars and Building Structures

When placing equipment in facilities that contain supporting structure, such as weight room pillars, columns, or other weight room obstructions, logistical planning for training sessions and safety factors should be prioritized. Due to the large size of many strength and conditioning facilities, the inclusion of weight-supporting and other building structures may be unavoidable. Strategically placing training equipment with consideration of any vertical weight room pillars or horizontal structural beams and designating specific exercise and storage areas within the facility layout considering proximity to building structures minimize facility congestion and will improve training session organization. Adhering to the safety spacing

and equipment guidelines in this chapter around building structures will improve the functionality and traffic flow of the facility.

Safety, function, and traffic flow are top priorities when deciding on the placement of equipment in a facility.

Stretching and Warm-Up Area

A stretching and warm-up area is an open area with soft tissue instruments, mats, or elastic bands. This area may include foam rollers and other tissue rolling tools (e.g., PVC pipes, and lacrosse, tennis, golf, or softball balls) and jump ropes. Mats can be used so that athletes do not have to lie on a hard floor, but this is not necessary.

Circuit Training Area

A circuit training area is typically a space with machines lined up or grouped together to make it easy for people to switch from one machine to the next. Grouping the machines can be done in several ways. Examples of different groupings include upper body and lower body, pushing actions and pulling actions, and various body part groupings. In many facilities, injured athletes use the circuit machines to exercise the body parts that are still healthy. Ease of access between machines should be kept in mind so that injured athletes can move freely. Stationary circuit training machines should be at least 30 inches (76 cm) (3), preferably 36 inches (91 cm), away from each other to provide a wide enough area and some extra space for safety. Safety spacing in the weight room is generally referred to as a **safety cushion**.

Free Weights

Free weight equipment includes dumbbells, barbells, benches, kettlebells, farmer's walk handles, hexagonal bars, squat racks, and any other equipment necessary to use these items. Kettlebells can be lined up neatly under a dumbbell rack or in their own section.

Since most activities with a kettlebell are dynamic and require space to complete a variety of movements, this section should be large enough to accommodate both the exercises being performed and the number of athletes using the space simultaneously.

Weightlifting Area

A weightlifting area generally contains racks with platforms or a platform standing alone. It could also be an open area where weightlifting is performed without a platform. This is typically done on a rubber floor with a concrete foundation. To keep things organized, weight trees or racks and bar holders can be used when the equipment is not in use.

Racks and platforms should have enough room between them that if someone were to fall, people nearby would not be injured. Weightlifting racks should be bolted to the floor to ensure that they do not move when in use. If a rack is portable, it should be moved to a designated storage area when not in use.

Exercise and Performance Testing Spaces

Exercise testing equipment should be in their own section or dedicated space. Exercise testing spaces are where equipment is grouped together for exercise testing sessions. This equipment may consist of force plates, isometric mid-thigh pulling racks, exercise testing bikes and treadmills, testing and monitoring device charging stations, and associated computer equipment. Ensuring a consistent location for exercise testing within the training facility is optimal for ensuring the validity and reliability of exercise testing.

It is also important to consider safety in implementing technology devices that are specifically designed for integration into free weight and weightlifting areas. Such equipment includes rack-mounted training feedback tablets, barbell attachments, linear position transducers, accelerometers, and mounted camera systems. As a general rule, weight room technology should be set up and administered in a way that does not impede planned safety space

(i.e., safety cushion) or interfere with the position of the spotters. Additional considerations should be made for the placement of clocks and television monitors that contain pertinent information to ensure optimal visibility throughout the training session.

Aerobic Area

The aerobic area is where the cardiovascular training equipment is grouped together. This equipment consists of stationary bicycles, stair climbers, elliptical trainers, treadmills, rowers, and so on. A general guideline is to allocate approximately 40 to 60 square feet (3.7-5.6 m²) of floor space per piece of equipment (2). Specific floor space requirements for aerobic equipment and the recommended space between pieces of equipment are provided in [table 25.1](#). These requirements outline the amount of space needed for each piece of equipment, along with the required amount of space to allow for room to move and prevent accidents caused by falling into equipment. Within aerobic training areas it is important that there is adequate spacing around and behind all continuous-motion exercise equipment (e.g., treadmills, elliptical trainers, bikes, ski trainers, and rowers) (2).

TABLE 25.1 Aerobic Equipment Space Estimates

Equipment	Minimum space needed
Bike (upright)	30 ft ² (2.8 m ²)
Bike (recumbent)	38 ft ² (3.5 m ²)
Elliptical trainer	47 ft ² (4.4 m ²)
Ski trainer	31 ft ² (2.9 m ²)
Stair climber	35 ft ² (3.3 m ²)
Rower	48 ft ² (4.5 m ²)

Adapted by permission from B. Kroll, "Structural and Functional Considerations in Designing the Facility: Part I," *National Strength and Conditioning Association Journal* 13, no. 1 (1991): 51-58, with updates made to comply with ADA Standards for Accessible Design (United States Department of Justice, 2010), 104-116, 224-252.

Equipment	Minimum space needed
Treadmill	53 ft ² (4.9 m ²)

Adapted by permission from B. Kroll, "Structural and Functional Considerations in Designing the Facility: Part I," *National Strength and Conditioning Association Journal* 13, no. 1 (1991): 51-58, with updates made to comply with ADA Standards for Accessible Design (United States Department of Justice, 2010), 104-116, 224-252.

Indoor Sprint Track and Turf Lanes

Indoor track and turf lanes are beneficial for incorporating plyometric, speed, and agility training. These surfaces also can be used to implement warm-up and dynamic movement programs with athletes. When track and turf lanes are incorporated within training facilities, in proximity to other strength and conditioning equipment, special considerations should be taken to enhance safety and reduce the risk of injury and liability concerns.

Based on the specific needs and training goals, sideline, distance, and running lane markings should be considered to optimize session management and athlete instruction. Track and turf lanes should remain clear of unnecessary equipment to avoid tripping and other injury. Considering how many athletes will be using the turf and track lanes simultaneously is essential to identifying space requirements.

The length of track and turf lanes will determine the ability to administer certain assessments. For example, a 20-meter (22 yd) track lane would not be suitable to perform the Yo-Yo intermittent recovery test due to the added 5 meters (5.5 yd) of length required for jogging recovery between test intervals. Also, the amount of space required at the ends of the track or turf lanes to allow for deceleration is an important safety consideration. Determining the amount of space needed to decelerate will depend on the demographic and skill level of those being trained. While it is common for padding to be installed on back-end walls to prevent injury, it is best to avoid unnecessary collisions with an adequate space allowance.

Suspension Training Equipment

Suspension training equipment refers to commercially available ring, handle, or strap setups (8). They are most often used for bodyweight training focusing on upper body, lower body, and core. The straps of a suspension training unit can be anchored independent of each other (e.g., gymnastics ring) or connected by a common anchor (e.g., eyelet and carabiner). Specific usage should be in accordance with setup and loading guidelines determined by the equipment manufacturer.

Safe and effective suspension training requires a stable attachment point, level nonskid flooring, open space free of obstacles, and handles that are easy to grip (e.g., not worn or slippery) (8). Because suspension training equipment must be attached to a fixed structure (e.g., load-bearing wall mount, bolted rack, or rig frame), it is suggested that the strength and conditioning professional regularly checks the integrity of the anchor point before each use. Ideally, the anchor point should be between 7 to 9 feet (2.1-2.7 m) above the floor. Further, suspension training equipment should not be mounted in or near doorways to prevent blocking emergency exits and hindering access to storage areas, cleaning closets, locker rooms, and restrooms.

All accessories used to mount suspension training equipment should be load bearing. As a general rule, any carabiner and snap hook used for exercise purposes should be load-rated for physical activity and include a secondary screw-locking mechanism to prevent risks of falls or drops causing a potential injury. In the United States, carabiners used for fall protection must meet OSHA standards, which specify a minimum tensile (breaking) strength of 5,000 lbs (22kN) (17). Outside of the United States, Conformité Européenne (CE) or other third-party rating requirements should be referenced to select connectors with appropriate load ratings for safety.

[Table 25.2](#) summarizes the space needs for a range of exercise types. Note that there are unique formulas that are specific to each

type with several example calculations, including those based on metric approximations.

TABLE 25.2 Calculations for Space Needs

Exercise type	Examples of exercises	Formulas and examples
Prone and supine exercises with bar	Bench press; lying triceps extension; hip thrust	Formula: DEPTH: actual weight bench length ^a (6-8 ft [1.8-2.4 m]) + a safety cushion of 3 ft (0.9 m) <i>multiplied by</i> WIDTH: actual bar length (4-7 ft [1.2-2.1 m]) + a double-wide safety cushion of 6 ft (1.8 m)
		Example 1: If using a 6 ft long weight bench and a 7 ft Olympic bar for the bench press exercise: DEPTH: (6 ft [bench] + 3 ft [safety cushion]) × WIDTH: (7 ft [bar] + 6 ft [double-wide safety cushion]) = 117 ft ²
		Example 2 (metric approximations): If using a 2 m long weight bench and a 1 m long bar for the lying triceps extension exercise: DEPTH: (2 m [bench] + 1 m [safety cushion]) × WIDTH: (1 m [bar] + 2 m [double-wide safety cushion]) = 9 m ²
Standing exercises	Biceps curl; upright row	Formula: DEPTH: a suggested user space of 2 ft (0.6 m) for a standing exercise + a safety cushion of 3 ft (0.9 m) <i>multiplied by</i> WIDTH: actual bar length ^b (4-7 ft [1.2-2.1 m]) + a double-wide safety cushion of 6 ft (1.8 m)
		Example 1: If using a 4 ft curl bar for the biceps curl exercise: DEPTH: (2 ft [user space] + 3 ft [safety cushion]) × WIDTH: (4 ft [bar] + 6 ft [double-wide safety cushion]) = 50 ft ²
		Example 2 (metric approximations): If using a 2 m bar for the upright row exercise: DEPTH: (1 m [user space] + 1 m [safety cushion]) × WIDTH: (2 m [bar] + 2 m [double-wide safety cushion]) = 8 m ²
Standing exercises from a rack	Back squat; shoulder press	Formula: DEPTH: a suggested user space of 8-10 ft (2.4-3.0 m) for a standing exercise (from a rack) <i>multiplied by</i> WIDTH: actual bar length (5-7 ft [1.5-2.1 m]) + a double-wide safety cushion of 6 ft (1.8 m)
		Example 1: If using a 7 ft Olympic bar for the back squat exercise: DEPTH: (10 ft [user space]) × WIDTH: (7 ft [bar]

Exercise type	Examples of exercises	Formulas and examples
		+ 6 ft [double-wide safety cushion]) = 130 ft² Example 2 (metric approximations): If using a 2 m Olympic bar for the shoulder press exercise: DEPTH: (3 m [user space]) × WIDTH: (2 m [bar] + 2 m [double-wide safety cushion]) = 12 m²
Weightlifting exercises	Power clean; power snatch	Formula (on inlaid rubber flooring): DEPTH: a suggested user space of 8-10 ft (2.4-3.0 m) for a weightlifting exercise <i>multiplied by</i> WIDTH: actual bar length (6-7 ft [1.8-2.1 m]) + a double-wide safety cushion of 6 ft (1.8 m) Formula (on an elevated wood platform): DEPTH: lifting platform length (typically 8 ft [2.4 m]) + a safety cushion of 3 ft (0.9 m) <i>multiplied by</i> WIDTH: lifting platform width (typically 8 ft [2.4 m]) + a double-wide safety cushion of 6 ft (1.8 m) Example 1 (on inlaid rubber flooring): DEPTH: (10 ft [user space]) × WIDTH: (7 ft [bar] + 6 ft [double-wide safety cushion]) = 130 ft² Example 2 (on an elevated wood platform; metric approximations): DEPTH: (2.5 m [platform length] + 1 m [safety cushion]) × WIDTH: (2.5 m [platform width] + 2 m [double-wide safety cushion]) ≈ 16 m²
Landmine exercises	Landmine squat; half-kneeling landmine press	Formula: The minimum space need is based on the area (A) of a circle ($A = \pi \times r^2$): The radius (r) of the circle is comprised of the actual bar length (5-7 ft [1.5-2.1 m]) + a suggested user space of 2 ft (0.6 m) for a landmine exercise + a safety cushion of 3 ft (0.9 m). A minimum of 90 degrees of movement is required for the range of motion at the user end of the bar for a landmine exercise; 90 degrees is one-quarter of a 360-degree circle. Thus, the formula is ($\pi \times [\text{actual bar length} + \text{user space} + \text{safety cushion}]^2$) <i>divided by 4</i>. (Use 3.14 for the value of π.) Example 1: If using a 7 ft Olympic bar for the landmine squat exercise: ($\pi \times [7 \text{ ft } \{\text{bar}\} + 2 \text{ ft } \{\text{user space}\} + 3 \text{ ft } \{\text{safety cushion}\}]^2$) ÷ 4 ≈ 113 ft²

Exercise type	Examples of exercises	Formulas and examples
		Example 2 (metric approximations): If using a 2 m Olympic bar for the half-kneeling landmine press exercise: $(n \times [2 \text{ m \{bar\}} + 1 \text{ m \{user space\}} + 1 \text{ m \{safety cushion\}}]^2) \div 4 \approx 12.5 \text{ m}^2$
Suspension training exercises	Suspension row; foot-elevated knee tuck	Formula: DEPTH: actual distance of the foot position (the farthest foot position if the feet are staggered) from the anchor point of the suspension strap (measure out from a spot on the ground directly under the anchor point) + the athlete's height + a safety cushion of 3 ft (0.9 m) multiplied by WIDTH: a suggested user space of 2 ft (0.6 m) for a suspension exercise + a double-wide safety cushion of 6 ft (1.8 m) Example 1 (angled use): If a 6 ft tall athlete is doing an angled suspension exercise (e.g., a suspension row) with both feet positioned 3 ft from the suspension strap anchor point: DEPTH: (3 ft [distance out from under the anchor point] + 6 ft [athlete's height] + 3 ft [perimeter walkway safety cushion]) $\times$ WIDTH: (2 ft [user space] + 6 ft [safety cushion]) = 96 ft² Example 2 (vertical or perpendicular use; metric approximations): If a 1.6 m tall athlete is doing a prone suspension exercise (e.g., a foot-elevated knee tuck) with the feet in the straps directly below the anchor point: DEPTH: (0 m [distance out from under the anchor point] + 1.6 m [athlete height] + 1 m [safety cushion]) $\times$ WIDTH: (1 m [user space] + 2 m [safety cushion]) $\approx 8 \text{ m}^2$

^aFor prone and supine ground-based barbell exercises with no bench, use the athlete's height for the DEPTH.

^bFor dumbbell and kettlebell exercises (i.e., standing exercises without a bar), use a user space of 2 ft (0.6 m) for the WIDTH.

Organizing a facility begins with arranging the equipment in a safe, functional space. The guidelines in this chapter provide the tools necessary to properly arrange the equipment. Taking a blank floor plan of the facility and sketching the equipment arrangement on the blank plan will help with equipment placement. [Figures 25.3](#) and [25.4](#) on [pages 812](#) and [813](#), respectively, show examples of

weight room floor plans for secondary school and university settings. Notice the walkway spacing and arrangement, as well as the organization and grouping of the equipment. Also note the placement of the supervisor's office, window placement, and storage areas.

Maintaining and Cleaning Surfaces and Equipment

Surfaces in the strength and conditioning facility should be cleaned, sanitized, and disinfected consistently to ensure that the equipment is safe and functional. Proper maintenance will reduce cost in the long run, because cleaning supplies are cheaper than replacing padding, flooring, exercise mats, and so on. Microbes will begin to grow if the surfaces are not wiped down with a germicidal cleaner. Equipment and surfaces that are used more often and have a greater chance of developing bacterial growth (e.g., padding and vinyl), should be cleaned daily or every other day. It is important that any germicidal cleaner used in the strength facility have the ability to prevent the spread of germs, including bacteria, viruses, parasites, and fungi, and illness, such as the flu, COVID-19, HIV, hepatitis, and ringworm (or other infectious rashes). After cleaning surfaces and equipment to remove any dirt and debris, the CDC recommends sanitizing and disinfecting surfaces with an EPA-registered disinfecting product to eliminate and inactivate germs to reduce the spread of illness (20). For laundry items, such as cleaning rags and gym towels, use detergent and the hottest available water setting, and dry items completely.

Nonabsorbent floors should be regularly mopped for removal of dust and grime buildup. Wood weightlifting platforms should also be checked for splintering and cracking and cleaned with an appropriate cleaner to remove dust and dirt that could cause foot slippage. When cleaning around racks and machinery, the bolts and screws that hold the equipment to the floor should also be tightened or checked regularly (4). Machines that have cables and pulleys should be regularly checked and should be repaired if the cables and pulleys are loose or fraying. If the facility has a tiled or rolled rubber floor, minimal gaps should appear between the pieces and no visible glue

should extrude between the cracks. Carpet should be vacuumed and cleaned regularly to prevent mold, mildew, or fungal growth.

Walls and ceilings should also be cleaned at least once every week or two. The walls and ceilings should have no dirt buildup, and there should not be any dust in the corners. Dust and cobwebs tend to build up high in the corners where the walls meet the ceiling and should be removed. If the facility has windows and mirrors, they should be regularly checked for cracking and replaced as soon as a crack is noticed. Clean windows and mirrors also enhance the appearance of the facility. Use a window cleaner and microfiber towel to leave the surface streak free. Dirt and dust can build up on flat edges of items affixed to the wall or fixtures hanging from the ceiling. Any windowsills, shelves, or anything hanging from the ceiling should be dusted regularly. If exercise equipment is hanging from the ceiling, it should be regularly checked to ensure that it is tightly secured and will not fall down (4). [Figure 25.5](#) on [page 814](#) presents a list to refer to when creating a cleaning schedule for the floors, walls, and ceilings.

Scheduling frequent maintenance and cleaning ensures safety, protects equipment, and maintains the appearance of the strength and conditioning facility.

Maintenance should include not only the surface materials but also the equipment. The equipment should be regularly checked for broken or damaged parts, particularly those that affect functionality. If the equipment is used often and not cleaned enough, residue can build up that may cause functional or health issues. Nonfunctional equipment should be labeled with an “Out of Order” sign (2). If there is a delay before fixing broken equipment, it should be removed from the floor and placed in storage. As with surfaces, equipment repairs can end up costing more if it is not maintained and cleaned properly. Malfunctioning equipment also increases the

risk of legal liability to strength and conditioning professionals and institutions (16).

The NSCA's safety checklist for exercise facility and equipment maintenance ([figure 25.6](#); [pages 815-817](#)) is helpful for determining the maintenance needs of exercise equipment. A cleaning schedule should be made that specifies the equipment to be cleaned daily, weekly, biweekly, or monthly. The checklist also covers facility layout issues, especially those relating to safety.

Cleaning and maintenance materials should be kept in a locked storage closet for use when needed. Any hazardous cleaning products and chemicals should be clearly labeled and only accessible to adults. Any tools should be stored in a toolbox that is put away and out of sight of the weight room. Tools and cleaning materials should also be regularly inventoried and restocked. A list of maintenance equipment and cleaning supplies is provided in [table 25.3](#).

TABLE 25.3 Equipment and Supplies for Maintenance and Cleaning

Maintenance equipment	Cleaning supplies
File	Disinfectant (germicide)
Hammer	Specialty cleaners (e.g., wood, walls, upholstery)
Pliers	Glass cleaner
Screwdriver set	Lubrication
Allen wrench set	Paper towels
Adjustable wrench set	Spray bottles
Knife, heavy-duty scissors, and/or box cutter	Cloth towels and rags
Stapler	Sponges
Duct tape	Brooms and dustpans
Extra nuts, bolts, and washers	Vacuum cleaner and/or wet vacuum cleaner
Heavy-duty glue	Mop and bucket
Drill and drill bit set	Stain remover
Vise grips	Wire brush
Adjustable ladder or step stool	Cleaning extension poles (for out-of-reach ceilings, mirrors, and walls)

Conclusion

Designing a strength and conditioning facility is an extensive process that involves a significant amount of planning. The process begins with the formation of a committee that will contribute to the design and construction of the facility. The next step involves assessing the specific needs of the strength and conditioning department. This is the phase in which the number and experience level of athletes being trained, preferences from strength and conditioning professionals and staff, scheduling, and equipment needs should be determined. After these determinations have been made, the process of designing the facility and arranging the equipment can begin. Care should be taken and guidelines followed in arranging and spacing equipment to optimize safety, function, and traffic flow.

After the design process is complete, the facility and the equipment must be properly maintained and cared for to ensure safe training and protect financial investments. Regular maintenance and cleaning of equipment and surfaces are vital for the longevity of the facility, in addition to health and safety. Cleaning and maintenance should focus the most on high-traffic areas and equipment that is used more often. This will help to prevent the spread of infections and disease among the facility users as well as promote proper hygiene and maintain the appearance of the environment. It is also suggested that once the facility is operational, key stakeholders meet periodically to discuss outcomes and plans for future changes (9).

KEY TERMS

- construction phase
- design phase
- feasibility study
- master plan
- minimally compliant facility
- needs analysis

predesign phase
preoperation phase
safety cushion
SWOT analysis

STUDY QUESTIONS

1. What is considered the optimal temperature range for a strength and conditioning facility?
 - a. 58 °F to 64 °F (14-18 °C)
 - b. 64 °F to 68 °F (18-20 °C)
 - c. 68 °F to 74 °F (20-23 °C)
 - d. 74 °F to 78 °F (23-26 °C)
2. What is the minimum recommended amount of space per person within a strength and conditioning facility?
 - a. 70 square feet (6.5 m²)
 - b. 100 square feet (9.3 m²)
 - c. 130 square feet (12.1 m²)
 - d. 144 square feet (13.4 m²)
3. What is the minimum recommended spacing on each side of a power rack?
 - a. 3 ft (0.9 m)
 - b. 4 ft (1.2 m)
 - c. 6 ft (1.8 m)
 - d. 7 ft (2.1 m)
4. What is the recommended width for designated walkways in a strength and conditioning facility?
 - a. 2 to 3 feet (0.6-0.9 m)
 - b. 3 to 5 feet (0.9-1.5 m)
 - c. 5 to 7 feet (1.5-2.1 m)
 - d. >7 feet (>2.1 m)
5. What is an important consideration for safely using suspension trainers?

- a. Regularly check that all suspension anchors, accessories, and attachments are properly secured for use with consideration of the load-bearing nature of the exercise.
- b. Finding a reinforced doorway to secure suspension training devices is best to optimize movement throughout a strength and conditioning facility.
- c. Single-lock carabiners must be used to attach suspension trainers to anchor points.
- d. Safety cushion spacing does not apply to suspension trainers.

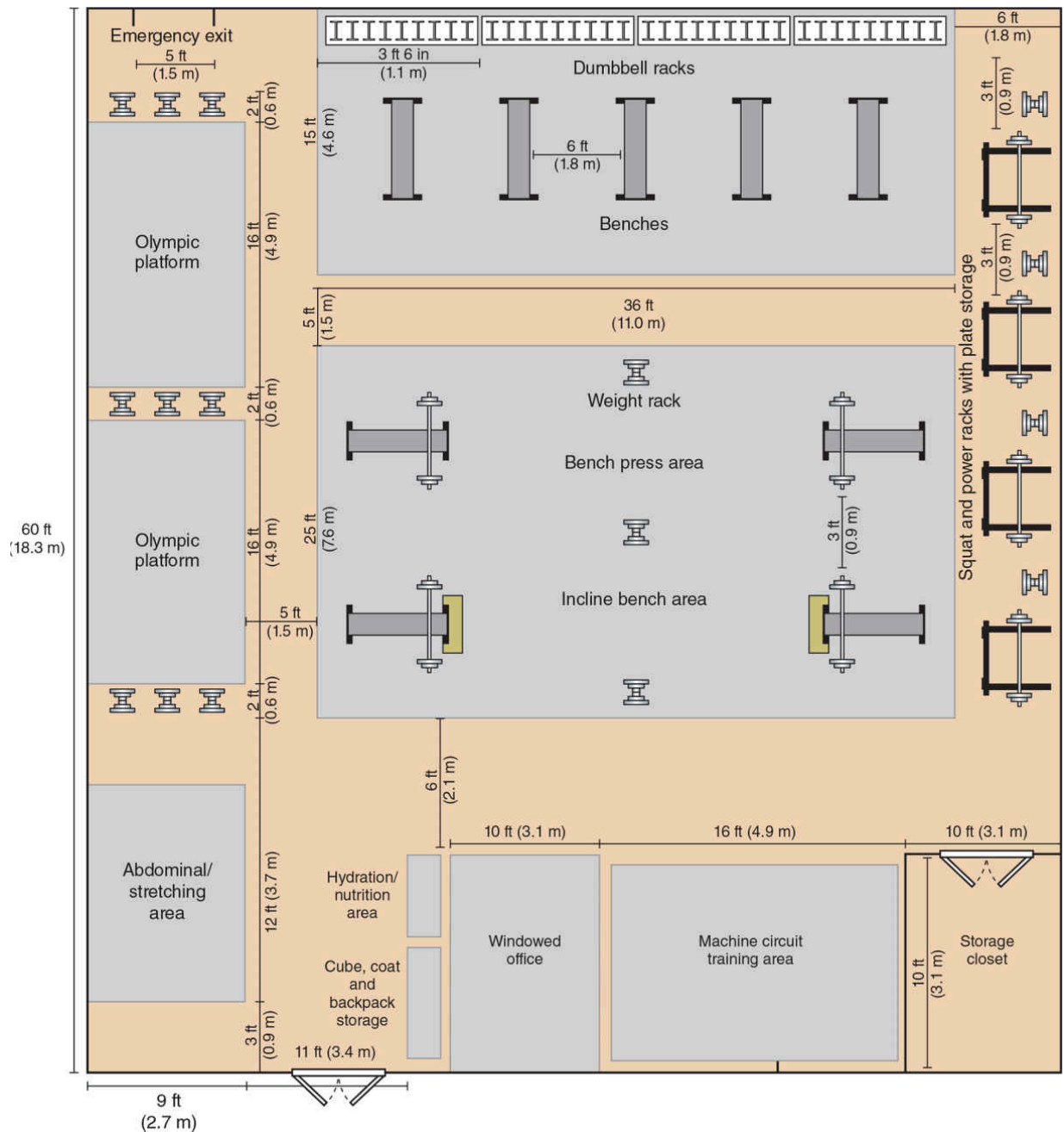


FIGURE 25.3 Example of a secondary school strength and conditioning facility floor plan (3 ft = 0.9 m).

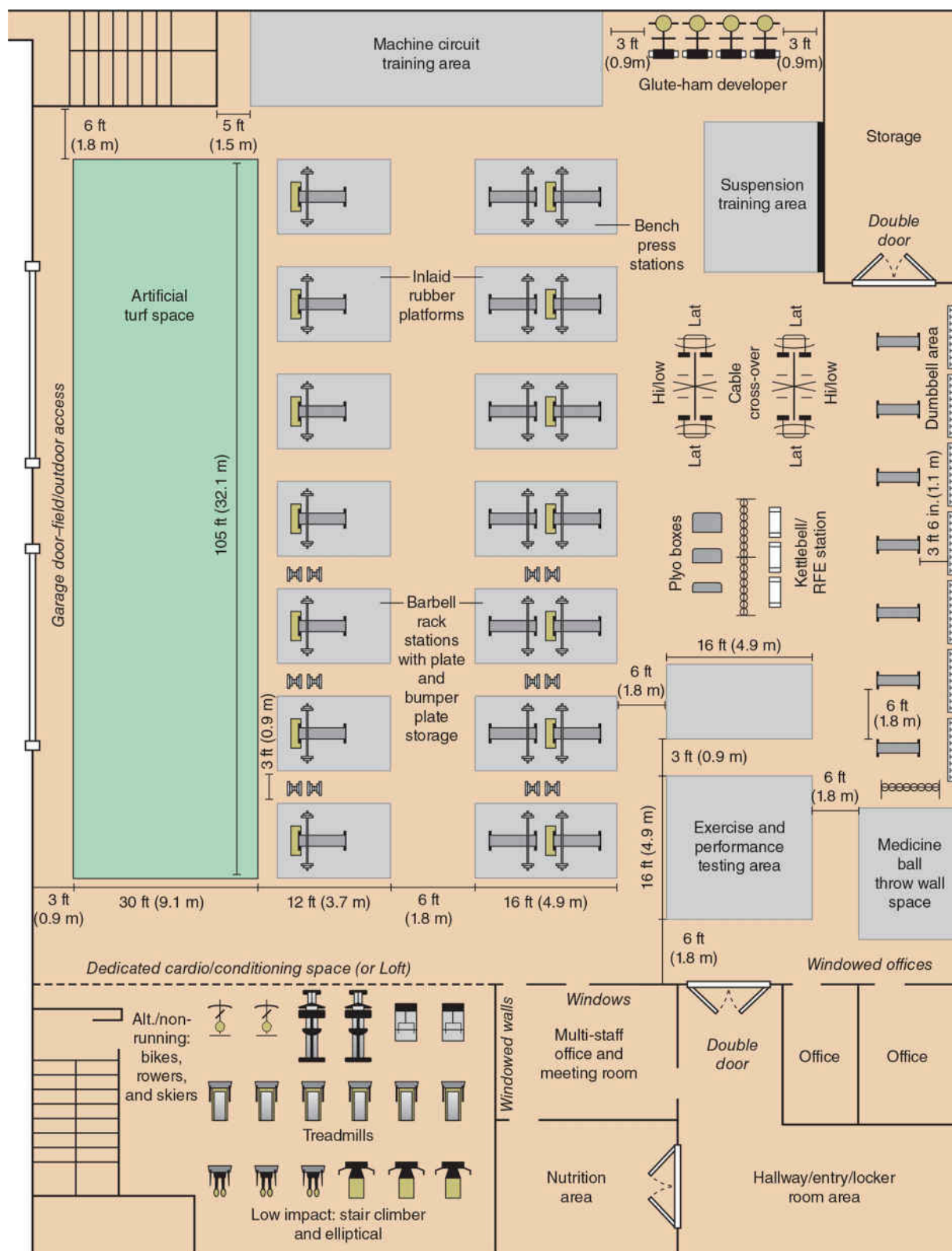


FIGURE 25.4 Example of a university strength and conditioning facility floor plan (3 ft = 0.9 m).

Figure 25.5 Checklist for cleaning floors, walls, ceilings, and surfaces.

Floors

- ☐ Check floors for dirt, grime, and chalk buildup.
- ☐ Nonabsorbent floors should be regularly swept and mopped.
- ☐ Check carpet for any moisture, mold, mildew, or fungal growth.
- ☐ Carpet should be vacuumed and cleaned regularly.

Walls

- ☐ Check walls for dirt and dust buildup.
- ☐ Dust and clean walls at least once every week or two.
- ☐ Clean mirrors and windows of smudges at least once per week.

Ceilings

- ☐ Ceilings and ceiling fixtures should be dusted or cleaned at least once every week or two.
- ☐ Open ceilings with exposed pipes and ducts should be cleaned as needed.
- ☐ Check for dust and cobweb buildup (especially near corners and HVAC vents).

Surfaces

- ☐ Dust windowsills and shelving surfaces at least once per week.
- ☐ Exercise surfaces, mats, and pads that contact skin should be cleaned daily with germicidal cleaner.
- ☐ Clean and sanitize hydration and nutrition spaces (drinking fountains, water bottle refill stations, countertops) daily and additionally as needed.

From National Strength and Conditioning Association, *Essentials of Strength Training and Conditioning*, 5th ed., edited by G.G. Haff and N.T. Triplett (Human Kinetics, 2027).

Figure 25.6 NSCA's safety checklist for exercise facility and equipment maintenance.

Exercise Facility

Flooring and Turf Areas

- ☐ Wooden platforms should be free of splinters, cracks, holes, protruding nails, loose screws, and warping.
- ☐ Tighten bolts and screws and ensure they are secure in attaching equipment to the floor.
- ☐ Tile flooring should be resistant to slipping; no moisture or chalk accumulation should appear.
- ☐ Rubber flooring should be slip resistant and free of cuts, slits, and gaps between pieces.
- ☐ Interlocking mats are secure and arranged with no protruding tabs.
- ☐ Carpet should be free of tears; any worn areas should be protected by mats.
- ☐ Flooring and turf are glued or fastened down properly.

Walls

- ☐ Walls in high-activity areas are free of protruding appliances, equipment, or wall hangings.
- ☐ Mirrors and shelves are securely fixed to walls.
- ☐ Mirrors are placed a minimum of 20 inches (51 cm) off the floor in all areas.
- ☐ Mirrors and windows are not cracked, broken, or distorted (replace immediately if damaged).
- ☐ Bolts and screws are tightened and secure in attaching any balance railing to the walls.
- ☐ Wall hooks and suspension anchors are secure for use with attachments.

Ceiling

- ☐ Ceiling fans, HVAC, and air exchange systems work properly.
- ☐ Lighting is designed to optimize visibility at floor level.
- ☐ Ceiling height is sufficient for overhead exercises (12 ft [3.7 m] minimum) and free of low-hanging apparatus (e.g., beams, pipes, lighting, signs).
- ☐ Ceiling fixtures and attachments are secure, preventing the risk of falling objects.
- ☐ Any damaged or missing ceiling tiles are replaced as soon as possible.

Exercise Equipment

Stretching and Bodyweight Exercise Area

- ☐ Mat area is free of weight benches and equipment.
- ☐ Mats and bench upholstery are free of cracks and tears.
- ☐ Equipment is properly stored after use.
- ☐ Elastic bands are secured to their base and checked for wear.

- ☐ Nonslip material appears on the top surface and bottom or base of plyometric boxes.

Resistance Training Machine Area

- ☐ Each station is easily accessed (a minimum of 30 in. [76 cm] between machines; 36 in. [91 cm] is optimal).
- ☐ Area is free of loose bolts, screws, cables, and chains.
- ☐ Proper selectorized pins are used and there is adequate storage or hanging space for equipment accessories.
- ☐ Securing straps and locking mechanisms must be functional.
- ☐ Functional parts are properly lubricated.
- ☐ Protective padding is free of cracks and tears.
- ☐ No protruding screws or parts need tightening or removed.
- ☐ Belts, chains, and cables are aligned with machine parts.
- ☐ No worn parts (e.g., frayed cable, loose chains, worn bolts, cracked joints) exist.

Resistance Training Free Weight Area

- ☐ Each bench or area is easily accessible (a minimum of 30 in. [76 cm] access clearance; 36 in. [91 cm] is optimal).
- ☐ Olympic bars are properly spaced (a minimum of 36 inches [91 cm] between ends).
- ☐ Dumbbell racks and other equipment are placed at least 6 inches (15 cm) away from mirrors.
- ☐ All equipment is returned after use to avoid obstruction of pathway.
- ☐ Safety equipment (e.g., belts, collars, safety bars) is used and returned.
- ☐ Protective padding is free of cracks and tears.
- ☐ Bolts and apparatus parts (e.g., collars and attachments) are secure and tightly fastened.
- ☐ Olympic bars turn properly and are properly lubricated and tightened.
- ☐ Benches, weight racks, standards, and similar equipment are secured to the floor or wall.
- ☐ Nonfunctional or broken equipment is removed from the area or labeled with an "out of service" sign.

Weightlifting Area

- ☐ Olympic bars are properly spaced (a minimum of 36 in. [91 cm]) between ends.
- ☐ All equipment is returned after use to avoid obstruction of lifting area.
- ☐ Olympic bars turn properly and are properly lubricated and tightened.

- ☐ Bent or rusted Olympic bars are replaced; knurling is brushed and clear of debris.
- ☐ Collars are functional.
- ☐ Bumper plates are inspected regularly; bumper plates should be free of cracks and any loose bolts.
- ☐ Sufficient grip chalk is available.
- ☐ Wrist straps, belts, and knee wraps are available, functional, and stored properly.
- ☐ Benches, chairs, and boxes are kept at a distance from weightlifting area.

Aerobic Exercise and Exercise Testing Areas

- ☐ Each station is easily accessible (minimum of 30 in. [76 cm] between machines; 36 in. [91 cm] is optimal).
- ☐ Bolts and screws are secure and tightened.
- ☐ Functioning parts are easily adjustable.
- ☐ Parts and surfaces are properly lubricated and cleaned.
- ☐ Foot and body straps are secure and not ripped.
- ☐ Recalibration and regular technology device maintenance are performed.
- ☐ Measurement devices are properly functioning.
- ☐ Power cords do not extend into exercise areas, safety spacing, walkways, or spotter areas.

Frequency of Maintenance and Cleaning Tasks

Daily

- ☐ Inspect all flooring and wooden platforms for damage or wear.
- ☐ Clean (sweep, vacuum, or mop and disinfect) all flooring.
- ☐ Inspect upholstery for damage or wear.
- ☐ Clean and disinfect upholstery.
- ☐ Clean and disinfect hydration and nutrition areas.
- ☐ Inspect fixed equipment's connection with floor, walls, or ceiling.
- ☐ Clean and disinfect equipment surfaces that contact skin.
- ☐ Inspect mirrors and windows for damage.
- ☐ Clean mirrors and windows of smudges.
- ☐ Inspect that wall hooks and suspension anchors are secure for use with attachments.
- ☐ Inspect all equipment for damage; wear; fraying; loose or protruding belts, screws, cables, or chains; insecure or nonfunctioning foot and body straps; improper functioning or improper use of attachments, pins, or other devices.

- ☐ Inspect all protective padding for cracks and tears.
- ☐ Inspect nonslip material and mats for proper placement, damage, and wear.
- ☐ Clean and sanitize exercise mats.
- ☐ Remove trash and garbage.
- ☐ Ensure equipment is returned and stored properly after use.

Two or Three Times Per Week

- ☐ Clean and lubricate all moving parts of equipment and aerobic machines.
- ☐ Clean and lubricate the guide rods on selectorized resistance training machines.
- ☐ Ensure Olympic bars turn properly and are properly lubricated and tightened.

Once Per Week or Two

- ☐ Clean and dust ceiling fixtures and attachments.
- ☐ Clean ceiling tiles.
- ☐ Clean walls.
- ☐ Clean light covers, fans, air vents, clocks, and speakers.

As Needed

- ☐ Replace light bulbs.
- ☐ Replace damaged or missing ceiling tiles.
- ☐ Clean open ceilings with exposed pipes or ducts.
- ☐ Remove or place “out of order” sign on broken equipment.
- ☐ Fill chalk boxes.
- ☐ Clean bar knurling.
- ☐ Remove rust from floor, plates, bars, and equipment with a rust-removing solution.
- ☐ Wash and sanitize rags and gym towels per use.

From National Strength and Conditioning Association, *Essentials of Strength Training and Conditioning*, 5th ed., edited by G.G. Haff and N.T. Triplett (Human Kinetics, 2027).